WORKED SOLUTION

STEP III 2022, Question 2

Pure •

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 1.5, *Infinite Descent*: https://beyondstep.keithmcnulty.org/ch01/infinite-descent.html

The key idea

This is Fermat's *method of infinite descent*. Suppose a solution in non-zero integers exists. A divisibility argument shows that every variable shares a common factor. Dividing it out gives a new solution of the same kind (or of a closely related kind) in strictly smaller non-zero integers. That can't go on for ever, because positive integers can't decrease for ever. In each part the work is to find the right prime to look at (2 or 3) and to check that the new solution really has the same form as the old one.

Solution

Throughout we use two facts. If a prime m divides x^k then m divides x. And if x is a non-zero multiple of $m \geq 2$, then x/m is a non-zero integer with $|x/m| < |x|$.

Part (i)

Suppose $a^3 + 2b^3 + 4c^3 = 0$ with a, b, c non-zero integers. Then

$$a^3 = -2\left(b^3 + 2c^3\right)$$

is even. The cube of an odd number is odd, so a is even. Write $a = 2p$. Then p is a non-zero integer with $|p| = \frac{1}{2}|a| < |a|$. Substituting,

$$8p^3 + 2b^3 + 4c^3 = 0 \quad \Longrightarrow \quad 4p^3 + b^3 + 2c^3 = 0.$$

This is the first claim.

Now repeat the argument on the new equation. It gives $b^3 = -2\left(2p^3 + c^3\right)$, so b is

even, say $b = 2q$, and then

$$4p^3 + 8q^3 + 2c^3 = 0 \quad \Longrightarrow \quad 2p^3 + 4q^3 + c^3 = 0.$$

Once more, $c^3 = -2\left(p^3 + 2q^3\right)$, so $c = 2r$, and

$$2p^3 + 4q^3 + 8r^3 = 0 \quad \Longrightarrow \quad p^3 + 2q^3 + 4r^3 = 0.$$

Here $p = a/2$, $q = b/2$ and $r = c/2$ are non-zero integers with $|p| < |a|$, $|q| < |b|$ and $|r| < |c|$, as required.

Ask yourself Why does this mean there is no solution at all, rather than just "another solution"?

Suppose solutions in non-zero integers exist. Among all of them, choose one for which $|a|$ is as small as possible. This is possible because every non-empty set of positive integers has a least element. The argument above then produces another solution (p, q, r) in non-zero integers with $|p| < |a|$, which contradicts the minimality of $|a|$. So no such integers a, b, c exist. (Equivalently, the process would give an infinite strictly decreasing sequence $|a| > |a/2| > |a/4| > \cdots$ of positive integers, which is impossible.)

Part (ii)

This time the prime to use is 3. Suppose $9a^3 + 10b^3 + 6c^3 = 0$ with a, b, c non-zero integers.

- Since $10b^3 = -3\left(3a^3 + 2c^3\right)$, 3 divides $10b^3$. As 3 does not divide 10, it divides b^3, and so it divides b. Write $b = 3q$. Then $9a^3 + 270q^3 + 6c^3 = 0$, that is,

$$3a^3 + 90q^3 + 2c^3 = 0.$$

- Now $2c^3 = -3\left(a^3 + 30q^3\right)$, so $3 \mid c$. Write $c = 3r$: $3a^3 + 90q^3 + 54r^3 = 0$, that is,

$$a^3 + 30q^3 + 18r^3 = 0.$$

- Finally $a^3 = -3\left(10q^3 + 6r^3\right)$, so $3 \mid a$. Write $a = 3p$: $27p^3 + 30q^3 + 18r^3 = 0$, that is,

$$9p^3 + 10q^3 + 6r^3 = 0.$$

So $(p, q, r) = (a/3, b/3, c/3)$ is a solution of the *same* equation in non-zero integers with $|p| < |a|$. As in part (i), choose a solution with $|a|$ minimal. This gives a contradiction, so there are no non-zero integer solutions.

Part (iii)

Any integer that is not a multiple of 3 can be written as $3n \pm 1$ for some integer n. Then

$$(3n \pm 1)^2 = 9n^2 \pm 6n + 1 = 3\left(3n^2 \pm 2n\right) + 1,$$

which is one more than a multiple of 3. A multiple of 3, on the other hand, has a square that is a multiple of 3 (indeed of 9).

So each square leaves remainder 0 or 1 on division by 3, with remainder 0 exactly when the number itself is a multiple of 3. For $x^2 + y^2$, the possible remainders are

$$0 + 0 = 0, \qquad 0 + 1 = 1, \qquad 1 + 1 = 2.$$

So $x^2 + y^2$ is a multiple of 3 only when both remainders are 0, that is, only when both x and y are multiples of 3.

Now suppose $a^2 + b^2 = 3c^2$ with a, b, c non-zero integers. The right-hand side is a multiple of 3, so by the result just proved $a = 3p$ and $b = 3q$. Then

$$9p^2 + 9q^2 = 3c^2 \quad \Longrightarrow \quad c^2 = 3\left(p^2 + q^2\right).$$

So c^2 is a multiple of 3. From the first result, c must then be a multiple of 3, since otherwise c^2 would be one more than a multiple of 3. Write $c = 3r$:

$$9r^2 = 3\left(p^2 + q^2\right) \quad \Longrightarrow \quad p^2 + q^2 = 3r^2.$$

This is a solution of the same equation in non-zero integers with $|p| < |a|$. Choosing a solution with $|a|$ minimal gives the same contradiction as before, so no non-zero solutions exist.

Part (iv)

Here we work modulo 4. An even number $2m$ has square $4m^2 \equiv 0$, and an odd number $2m + 1$ has square $4m^2 + 4m + 1 \equiv 1 \pmod 4$. So a sum of three squares leaves remainder equal to the number of odd terms among the three (0, 1, 2 or 3). It is a multiple of 4 only when all three numbers are even.

Ask yourself If we halve everything in $a^2 + b^2 + c^2 = 4abc$, do we get the same equation back?

No. If $a = 2p$, $b = 2q$ and $c = 2r$, the equation becomes $4(p^2 + q^2 + r^2) = 32pqr$, that is, $p^2 + q^2 + r^2 = 8pqr$. The coefficient has doubled. So we prove something stronger, which *is* preserved by halving:

Claim. For no integer $k \geq 2$ are there non-zero integers with $a^2 + b^2 + c^2 = 2^k abc$.

Suppose there were. Among all such solutions (with any $k \geq 2$), choose one with $|a|$ as small as possible. Since $k \geq 2$, the right-hand side $2^k abc$ is a multiple of 4. So $a^2 + b^2 + c^2$ is a multiple of 4, and by the remark above a, b and c are all even: $a = 2p$, $b = 2q$, $c = 2r$. Then

$$4\left(p^2 + q^2 + r^2\right) = 2^k \cdot 8pqr \quad \Longrightarrow \quad p^2 + q^2 + r^2 = 2^{k+1}\, pqr.$$

This is a solution in non-zero integers with exponent $k + 1 \geq 2$ and $|p| < |a|$, which contradicts the choice of a. So the claim holds. The case $k = 2$ shows that there are no non-zero integers a, b, c with $a^2 + b^2 + c^2 = 4abc$.

Checks

- A computer search over all non-zero a, b, c with $|a|, |b|, |c| \leq 40$ found no solution of any of the four equations, nor of $a^2 + b^2 + c^2 = 2^k abc$ for $2 \leq k \leq 6$.
- The power of 2 in the coefficient is what makes part (iv) work. Some equations of this shape *do* have solutions. For example, $1^2 + 1^2 + 1^2 = 3 \cdot 1 \cdot 1 \cdot 1$ and $3^2 + 3^2 + 3^2 = 1 \cdot 3 \cdot 3 \cdot 3$. There the coefficient is odd, so the mod-4 argument has nothing to bite on.
- Try the descent on a near miss: $a^2 + b^2 = 3c^2$ fails for $(a, b, c) = (1, 1, 1)$ because $1 + 1 = 2$ is not a multiple of 3, which is exactly the "$1 + 1 = 2$" case in part (iii).

Common slips

- Forgetting to say why the descent gives a contradiction. "We get smaller solutions" is not a proof until you invoke a minimal solution (or the fact that positive integers can't decrease for ever).
- In part (iv), halving and claiming the equation is unchanged. It isn't: the coefficient doubles, which is why the statement has to be strengthened.
- Writing "$3 \mid c^2$ so $3 \mid c$" without a reason. Here the reason is the first result of part (iii) (more generally, 3 is prime).

Beyond STEP

Infinite descent is how Fermat proved that $x^4 + y^4 = z^2$ has no solutions in positive integers, and so settled the $n = 4$ case of his Last Theorem. Each descent here amounts to showing that the equation has no solution modulo a suitable prime power in which the variables are not all divisible by that prime. This is the first step towards the p-adic numbers and the local–global ideas of modern number theory. The famous $a^2 + b^2 + c^2 = 3abc$ (the Markov equation) has infinitely many solutions, generated by a beautiful tree structure. Davenport's *The Higher Arithmetic* covers descent, congruences and sums of squares at exactly this level.

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